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A Novel Minimal Arabic Script for Preparing Databases and Benchmarks for Arabic Text Recognition Research

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Abstract: - This work presents a minimal Arabic script that may be used in training and testing of Arabic text recognition systems. Collecting handwritten samples from different writers to build handwritten text databases which may be used for benchmarking Arabic text recognition systems is another application. The suggested Arabic script covers the different shapes of Arabic alphabet in all positions (viz. standalone, initial, medial, and terminal). The frequency of each shape in the minimal text is designed to be the minimal possible. The suggested script is novel from different perspectives. A writer may participate with only three lines of meaningful Arabic text to cover all possible alphabet shapes, a total of 125 shapes. Collecting scripts from different writers provide evenly distributed letter frequencies that assure enough samples for all character shapes. This enables proper training resulting in more accuracy in the recognition phase. The same can be stated for printed Arabic text. This is especially useful when using large number of features with classifiers that require large number of samples for each category. Hidden Markov Models and Neural networks are two examples of these classifiers. In addition, the paper presents statistical analysis of Arabic corpora for estimating the number of occurrences of different shapes of Arabic alphabets in large corpora. The frequency of Arabic alphabet usage was utilized in enhancing the search for the minimal Arabic text.

Key-Words: - Arabic text recognition, Arabic OCR databases, Minimal Arabic script.

1 Introduction

This paper addresses a need for Arabic text recognition research as noted by several researchers [1-3]. There is a lack of Arabic text databases (both typed and hand-written) for use in Arabic text recognition system (ATRS) research. Such databases should include printed Arabic text with several types and sizes of fonts as well as handwritten text written by different writers. This need is addressed in this paper in two ways; by presenting a minimal text that covers all the shapes of all Arabic alphabet and by estimating the frequencies of the use of Arabic alphabet in different shapes and positions. The frequencies are estimated by statistically analyzing Arabic documents of over 4 million characters. This combination (viz. the minimal Arabic text and the frequencies of Arabic alphabets in corpora of Arabic text) may be used as alternative to the use of text databases in the research of Arabic text recognition.

Few Arabic databases with limited content are available for research in ATRS. Some of them have been prepared for specific domains and applications such as checks, numerals contents, and postal addresses. Arabic literal amounts of 4800 words were used in [4]. A database consisting of 26,459 Arabic names, presenting 937 Tunisian town/village names, handwritten by 411 different writers is presented in [5, 6] and used in several research work [7, 8]. A database prepared from text of 100 writers each wrote

67 literal numbers, 29 most popular words in Arabic writing, three sentences representing numbers and quantities used in checks, and a free subject chosen by the writer (around 4700 handwritten words produced by 100 writers.) is discussed in [9-12]. A small database for digits that involves 17 writers each wrote 10 digits 10 times each is presented in reference [13]. Arabic and Persian isolated characters database consisting of 220,000 handwritten forms filled by more than 50,000 writers with isolated characters is presented in [14]. The database in [15, 16] presents 29,498 images of sub-words, 15,175 images of Indian digits and image samples of each of legal and courtesy amounts from 3000 real-life bank checks. Another database for bank-checks includes 70 words of Arabic literal amounts extracted from 5000 checks by 100 writers was used in [17, 18]. An automatically generated printed database of 946 Tunisian town names is discussed in [19]. 360 handwritten addresses of around 4000 words were used in [20]. [21] Presents a database consisting of more than 17820 names of 198 cities of Iran. A general database with signatures has 37,000 words, 10,000 digits, 2,500 signatures, and 500 free-form Arabic sentences [22]. A small isolated character database consisting of 50 images for each character by 5 writers; each wrote the whole 28-character alphabet ten times was used in [23]. DARPA Arabic Corpus consists of 345 scanned pages of printed text in 4 different fonts [24]. The system in [25] was based on 40 pages of DARPA database. The research presented in [26] used 700 digitized pages from 45 printed documents. The segmentation work in [27] was based on around 240 digitized pages written by 178 writers, where each writer wrote one or two pages of 10 prepared pages of 13 lines each.

By reviewing the available databases for Arabic text recognition we found out that none of the databases contain even distribution of character frequencies of the collected data. In some cases, some characters are appearing 50 times or more than other characters (see [27] as an example). Moreover, if a handwritten database is targeted, it will be very hard to require writers to write long text, say one page or more. Even if we are able to collect two handwritten pages or more per writer some of the characters may not be present in the text with adequate frequency. In a test for training of a Hidden Markov Model using an arbitrary 2500 lines of Arabic text some characters had as low as 8 occurrences while other characters had over 1000 occurrences. It was noticed that characters with low number of occurrences had low correct recognition rates as expected due to the lack of enough samples for training. These reasons are some of the motivations that encouraged us to look for a minimal Arabic script to be used for preparing databases and benchmarks for Arabic text recognition research especially for handwritten text. To our knowledge this is the first attempt at addressing this problem using minimal text and hence no references are available for citation on minimal text coverage of Arabic language.

Table 1. Shapes of Arabic alphabets

no	S-alone	Term.	Medial	Initial	Shapes	Class
1	ء	ء	ء	ء	1	1
2	آ	ـآ	ـآ	آ	2	2
3	أ	ـأ	ـأ	أ	2	2
4	ؤ	ـؤ	ـؤ	ؤ	2	2
5	إ	ـإ	ـإ	إ	2	2
6	ئ	ـئ	ـئ	ئ	4	3
7	ا	ـا	ـا	ا	2	2
8	ب	ـب	ـبـ	بـ	4	3
9	ة	ـة	ـة	ة	2	2
10	ت	ـت	ـتـ	تـ	4	3
11	ث	ـث	ـثـ	ثـ	4	3
12	ج	ـج	ـجـ	جـ	4	3
13	ح	ـح	ـحـ	حـ	4	3
14	خ	ـخ	ـخـ	خـ	4	3
15	د	ـد	ـد	د	2	2
16	ذ	ـذ	ـذ	ذ	2	2
17	ر	ـر	ـر	ر	2	2
18	ز	ـز	ـز	ز	2	2
19	س	ـس	ـسـ	سـ	4	3
20	ش	ـش	ـشـ	شـ	4	3
21	ص	ـص	ـصـ	صـ	4	3
22	ض	ـض	ـضـ	ضـ	4	3
23	ط	ـط	ـطـ	طـ	4	3
24	ظ	ـظ	ـظـ	ظـ	4	3
25	ع	ـع	ـعـ	عـ	4	3
26	غ	ـغ	ـغـ	غـ	4	3
27	ف	ـف	ـفـ	فـ	4	3
28	ق	ـق	ـقـ	قـ	4	3
29	ك	ـك	ـكـ	كـ	4	3
30	ل	ـل	ـلـ	لـ	4	3
31	م	ـم	ـمـ	مـ	4	3
32	ن	ـن	ـنـ	نـ	4	3
33	ه	ـه	ـهـ	هـ	4	3
34	و	ـو	ـو	و	2	2
35	لآ	ـلآ	ـلآ	لآ	2	2
36	لأ	ـلأ	ـلأ	لأ	2	2
37	لإ	ـلإ	ـلإ	لإ	2	2
38	لا	ـلا	ـلا	لا	2	2
39	ى	ـى	ـى	ى	2	2
40	ي	ـي	ـيـ	يـ	4	3

It is clear from the above literature review that there is no adequate Arabic text databases freely available for use in the research of Arabic handwritten (and typewritten) text. The minimal text of this paper in addition to the work of the authors presented in [28], which presented the probabilities of the occurrence of

the Arabic alphabets in different positions of Arabic words, is an efficient solution to the above problem.
This paper is organised as follows: the characteristics of Arabic text are described in Section 2. The process of finding the minimal Arabic script is discussed in Section 3. The implementation of the system and experimental results are reported in Section 4. Finally, the concluding remarks are given in Section 5.

2 Some Characteristics of Arabic Text

Arabic is a cursive language written from right to left. It has 28 basic alphabets. An Arabic letter might have up to four different shapes depending on the position of the letter in the word: whether it is a standalone letter, connected only from right (initial form), connected only from left (terminal form), or connected from both sides (medial form). Letters of a word may overlap vertically (even without touching). Arabic letters do not have fixed size (height and width). Letters in a word can have diacritics (short vowels) such as *Fat-hah, Dhammah, Shaddah, sukoon* and *Kasrah*. Moreover, *Tanween* may be formed by having double *Fat-hah*, *Dhammah*, or *Kasrah.* Fig 1 lists these diacritics. These diacritics are written as strokes, placed either on top of, or below, the letters. A different diacritic on a letter may change the meaning of a word. Readers of Arabic are accustomed to reading un-vocalized text by deducing the meaning

Fat-hah ـَ Dhammah ـُ Shaddah ـّ
Kasrah ـِ Sukoon ـْ TanweenFat-h ـً
Tanween Dhamm ـٌ Tanween Kasr ـٍ

Fig 1. Arabic short vowels (diacritics)

from context. Fig 2 shows some of the characteristics of Arabic text. It shows a base line, overlapping letters, diacritics, two shapes of *Noon* character (initial

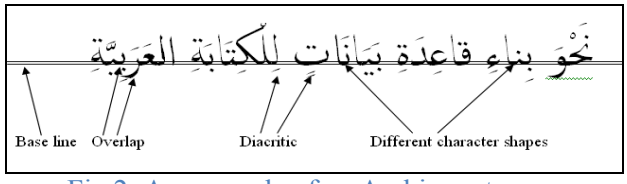


Fig 2. An example of an Arabic sentence indicating some characteristics of Arabic text.

and medial).

As Arabic numbers are not connected and are used globally, we concentrated our work on Arabic letters throughout this paper. As we stated earlier, Arabic has 28 main letters as shown in Fig 3. When considering presenting Arabic characters in computers, some of the main letters had been extended into separate letters for easiness of presentations and usability by the Arab Standardization and Metrology Organization (ASMO). The standard Arabic codepages (character sets) ASMO-449, ASMO-708 and ISO 8859-6 define 36 Arabic letters (See Fig 4). When we consider Arabic optical character recognition, we need to add *Lam-Alef* in its 4 different forms. Although *Lam-Alef* is a sequence of two alphabets, they are written as one set. We think that we need to treat this sequence as one set. So, we added four more sets to the alphabets; one with bare *Alef*, the second with *Alef-Maddah*, the third with *Alef-up-Hamza* and the fourth with *Alef-down-Hamza* as shown in Fig 5. This leads us to expand the number of Arabic letters to 40. Each alphabet can take different number of shapes (from 1 to 4). Hence, the total number of shapes is 125 (one letter has only one shape, others have two, and the most have four shapes).
Table 1 shows the basic Arabic letters with their categories. We are grouping them into 3 different classes according to the number of shapes a letter takes. The first Class (class 1) consists of a single shape of the *Hamza* which comes in stand-alone state (Number 1 in Table 1). *Hamza* does not connect with any other letter. The second class (class 2) presents the letters that can come either standalone or connected only from right (medial category). This class consists of *Alef Madda, Alef up Hamza, Waw*

ا ب ت ث ج ح خ د ذ ر ز س ش ص ض ط ظ ع غ ف ق ك ل م
ن ه و ي

Fig 3. Basic Arabic 28 letters

Hamza, Alef down Hamza, Alef, Tah Marboutah, Dal, Dhal, Ra, Zain, Waw, Lam Alef Madda, Lam Alef Hamza up, Lam Alef Hamza down, and Lam Alef (numbers 2-5, 7, 9, 15-18, and 35-39 in Table 1). The third class (class 4) consists of the letters that can be connected from either side or both sides as well as they can appear as standalone. This class consists of *Hamza Kursi, Baa, Taa, Thaa, Jeem Haa, Khaa, Seen, Sheen, Sad, Dhad, Dhaa, THaa, Ain, Gain, Faa, Qaaf, Kaaf, Laam ,Meem, Noun, Haa, Yaa* (numbers 6, 8, 10-14, 19-33, and 40 in Table 1). Table 2 shows a summary of these classes.

Although an Arabic letter might have up to 4

ء آ أ ؤ إ ئ ا ب ة ت ث ج ح خ د ذ ر ز س ش ص ض ط ظ ع غ
ف ق ك ل م ن ه و ى ي

Fig 4. Extend Arabic letters by ASMO.

different shapes, each letter is saved using only one code. It is the duty of a computer built-in driver to make contextual analysis to decide the right shape to

display, depending on the previous and next characters if available.

When it is needed to consider different shapes of Arabic letters for a given Arabic text file, a contextual analysis algorithm is needed. Such algorithm takes the letter, its predecessor, and its successor and identifies the right shape depending on the classes of the letters. This algorithm was implemented in this work to label characters to their positional shapes.

3 The Used Corpora

The used corpora for our analysis consist of Arabic

ء آ أ ؤ إ ئ ا ب ة ت ث ج ح خ د ذ ر ز س ش ص ض ط ظ ع غ
ف ق ك ل م ن ه و لآ لأ لإ لا ى ي

Fig 5. Expanded Arabic alphabets by adding different versions of Lam-Alef sequence.

text of two Arabic lexicons [29, 30], two *HADITH* books [31, 32] and a lexicon containing the meaning of Quran tokens in Arabic [33]. The electronic versions of such books and other old Arabic classical books could be found from different sites in the internet including [34].

4 The Minimal Arabic Script

The novel idea, which we are introducing here, is to use a script that consists of minimum number of letters (using meaningful Arabic words) that covers all possible shapes when preparing databases and benchmarks for Arabic optical character recognition. Several utility programs, implementing different

Table 1. Classes of Arabic alphabets depending on number of possible shapes.

Class	# of possible shapes	Alphabets
1	1	ء
2	2	آ أ ؤ إ ا ة د ذ ر ز و لآ لأ لإ لا ى
3	4	ئ ب ت ث ج ح خ س ش ص ض ط ظ ع غ ف ق ك ل م ن ه ي

algorithms to address this issue, were written to search huge corpora of Arabic script to find the minimum meaningful words that cover all Arabic alphabet-shapes.

As we stated earlier, Arabic text is saved using a unique code for each character irrespective of its position and shape. When statistics of Arabic character shapes are required, a procedure needs to be used to figure out the shape of each character. An algorithm was implemented to decode to tag the letters with their position code in the word according to the position of the letter in the word (initial. medial, terminal, or isolated). The classes used in the algorithm are those that are presented earlier and summarized in Table 2.

The following is a summary of the main steps of the simple algorithm we have used to process words extracted from Arabic corpora to generate the minimal text.

1. Initially the word is validated to see if it is already in the minimal text, if this is true then the word is not processed and the function is terminated.
2. The word is decoded to give the proper letter shapes of the word using the implemented contextual analysis algorithm.
3. The word is validated for multiple occurrences of a letter, if it has multiple occurrences of a letter then the word is not processed and the function is terminated as multiple occurrences are not allowed.
4. Each letter of the word is checked with the letters' table (holding the different shapes of Arabic alphabet). If any letter in the word is already flagged in the letters' table then the word is ignored and the function is terminated.
5. If a word passes the previous validations then
   a. The word is added to the minimal text, and
   b. The shapes of the all-shapes table corresponding to the letters of the word are flagged.

Several search criteria was conducted on the corpora

Table 3. Minimal text usage of the different shapes of Arabic alphabets

Letter	S-alone	Term	Initial	Medial
ث خ ش ض ظ غ ى	1	1	1	1
ئ ب ت ج ح س ص ط ع ف ق ك ل م ن ي	1	1	2	1
آ أ ؤ إ ا ة د ذ ر ز ه و لآ لأ لإ لا	1	1		
ء	1			

to generate the minimal Arabic text using this *algorithm*. In one criterion the corpora were searched sequentially for words from the beginning taking each word into account. This process was continued until the whole corpora are searched. Then the resulting letters' table is checked for un-flagged letters. It was clear that this process could not flag all shapes and hence the minimum text was not covering all shapes. In other criteria, different sequence of the data in the corpora was selected (i.e. the sequence of searching the corpora was changed several times). This did not result in acquiring a minimum text to cover all Arabic alphabet shapes. In another criterion the words are

randomly selected from the corpora. This showed better results, however, the minimal produced text was not including all Arabic alphabet shapes. Another search algorithm was executed which starts by selecting words having letters of minimal frequencies of usage utilizing the estimated frequencies of Arabic alphabet in Arabic script which is presented in Section 5. Hence, less frequently used letters were given higher priority. This resulted in improving the minimal produced text. In all of these experiments there was a constraint of not using a letter shape more than one time. It is to be noted that several versions of the algorithm were tested in this work.

The table of Arabic alphabets (Table 1) was analyzed and it showed that there are 39 shapes of letters that might come at the end of a word (terminal form) and 23 shapes of the letters that might come at the beginning of a word (initial form). Hence, there should be some repetitions of the letters shapes that come at the beginning. In order to include all the shapes of Arabic letters the previous search algorithms were applied again, allowing the possibility of an initial beginning letter to have up to

جعثق سؤق ذم ظأب رث مجس غضبى قمين شتف وس ضغط أي
بحظل حذف نسكه طخفة خصهم صئك إظ فت زؤى كنب دك آخ ال
هوج ثأئي طء لطاع يقره عزة تشدح لاغ لأص لآت لإض ش ئ مج
حث جح صخ فإن يجئ نص قش عض لظ بلغ سع

Fig 6. An Early minimal Arabic Script

two occurrences. In addition to these constraints, we limited the total number of extra occurrences of these letters to 16. By using corpora of around 20 Megabytes of text and running programs for several days, we could reach to a nearly optimal script. An early minimal script has been identified as shown in Fig 6. The script then has been optimized manually in several iterations until reaching the presented text in Fig 7.

The statistics of letter shape distribution for the suggested minimal Arabic script is shown in Table 3. It is clear from the table that 16 initial letter shapes have two occurrences each to compensate for the extra shapes of Arabic letter shapes that come at the end of a word (terminal). All other letter shapes are used only once. Hence, the presented text is the minimal possible text that covers all the shapes of Arabic alphabets. It is minimal in terms of the number of shapes used.

It might be clear to the reader that the minimal script is not unique. Theoretically speaking, there is infinite number of different scripts. A main characteristic of all these minimal scripts is that they all should have only 141 Arabic alphabet shapes that cover all Arabic letter shapes.

5 Arabic Script Statistics

Samples of classical Arabic corpora have been analyzed to find the distribution of basic Arabic letter shapes used in Arabic Script. A text of 4.25 million characters has been used. The text consists of Two *HADITH* books, *Al-Bukari* and *Muslim*. Table 4 presents the frequencies of using the different forms of Arabic letters in the two books of *Bukhari* and

عزة كأب جنة طفق ميس غضبى كثف ضغط أص فظل حذف نسكه
خصتهم صئك إظ رؤى شعب دك آخ ملأ سطوع يقرء تشدح لاغ للآت
لإض هج حث جح صخ فإني يجئ نخص قش عض تحظ بلغ سع ظمآن
قن طائي ثلاث لأج لآه بؤس ذم ائت للإوز ق ط ش ل ئ

Fig 7. The Minimal Arabic Script

Muslim. The frequencies of usage of Arabic may be biased by the type of text being searched. However, the frequency of usage of Arabic letter shapes of different books gave close results. The researchers will extend the search to more data in the future. We could have domain specific alphabet usage frequencies in addition to the general frequencies. As an alternative, previous work of the researchers [28] which addressed this issue by giving the probabilities of the usage of each letter in the different positions of Arabic words may be used.

6 Conclusion and Future Work

This paper presented minimal Arabic text that covers all of the Arabic letters with all possible shapes for each letter. We showed that the text is minimal and covers all the Arabic letter shapes. This presents one of the minimal solutions as this solution is not unique. The number of used letter shapes is the minimum possible to cover all the shapes of Arabic alphabet (i.e. 141 letters). However, different words may be generated as a minimal text depending on the algorithm of searching the corpora. Any new analysis may produce new words in the new minimal text. However, the number of letters in the minimal text that covers all Arabic alphabets cannot be less than the size of our minimal text (i.e. 141). Using this text we can build Arabic handwritten or typewritten Arabic databases, for use in Arabic text recognition systems, with minimal effort. In addition, we presented the frequencies of using the different shapes of Arabic alphabets in a data consisting of over 4 million characters. In our future work we will build

databases using this script and the probabilities of each Arabic letter in practical applications using our previous work related to the probabilities of Arabic alphabet usage. As we intend to use machine learning algorithms to recognise Arabic alphabet in the future, we believe that including the probabilities of each Arabic letter shape as one of the input features for the training stage will prove to be valuable. This database is expected to address one of the problems of conducting Arabic text recognition research (i.e. the lack of suitable Arabic text databases) as reported by many researchers.

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Table 4. Alphabet shapes distribution in classic Arabic For *Al-Bukari* and *Muslim*.

Let.	S-alone	Term.	Initial	Medial	Total
ء	11896				11896
ا	213359	296148			509507
إ	29670	6321			35991
أ	103938	22656			126594
آ	3551	1490			5041
ب	15541	9922	140434	67938	233835
ة	17597	37078			54675
ت	6132	27033	29353	35826	98344
ث	2261	4368	49989	8749	65367
ج	2964	1496	23817	13394	41671
ح	3258	3388	69860	31432	107938
خ	243	264	22807	7089	30403
د	18950	114451			133401
ذ	13526	15441			28967
ر	56138	115896			172034
ز	8623	12608			21231
س	5836	7233	75992	25487	114548
ش	330	1647	15469	13647	31093
ص	481	896	32115	13835	47327
ض	1562	1481	8791	6677	18511
ط	414	1170	4504	10245	16333
ظ	40	955	925	2599	4519
ع	1963	11170	136499	54625	204257
غ	217	483	5536	4608	10844
ف	2334	3655	76296	18397	100682
ق	4966	3497	64630	36482	109575
ك	3076	13896	29424	20548	66944
ل	68146	26147	242196	196946	533435
م	14831	62246	80246	80934	238257
ن	41669	130747	49601	103031	325048
ه	10781	122380	33486	37409	204056
و	111734	81885			193619
ؤ	792	2310			3102
ى	2870	62266			65136
ي	9800	72091	76211	120189	278291
ئ	184	184	6718	2852	9938
Total	789673	1274899	1274899	912939	4252410

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